2023-2024 Accomplishments

Journal Articles

- Lawrence V. Stanislawski, Barry J. Kronenfeld, Barbara P. Buttenfield, Ethan J. Shavers, 2023, At what scales does a river meander? Scale-specific sinuosity (S3) metric for quantifying stream meander size distribution, Geomorphology, Volume 436, 2023, https://doi.org/10.1016/j.geomorph.2023.108734.
- Costabile, P., Costanzo, C., Lombardo, M., Shavers, E., and Stanislawski, L., Unravelling spatial heterogeneity of inundation pattern domains for 2D analysis of fluvial landscapes and drainage networks. J. Hydrology, Volume 632, 2024, https://doi.org/10.1016/j.jhydrol.2024.130728

Community for Data Integration Project

- Stanislawski, L.V., Kronenfeld, B.J., Buttenfield, B.P., Shavers, E.J., and Marken, W.J., 2023. Generalization quality assessment tools (GQAT) for geospatial data.

Symposiums

- 2024 Symposium on Geospatial Data Science for Sustainability: Multiscale Mapping that Works: Harnessing Intelligent Methods for Cartographic Display and Analysis organized by B.J. Kronenfeld, Stanislawski, L.V., Buttenfield, B.P., Wang, S., Shavers, E.J., Kang, Y., Michels, A., Park, J. Co-organized with International Cartography Association Commission on Multiscale Cartography.

Presentations

- Stanislawski, L. Costabile, P., Costanzo, C., Duffy, A., Shavers, E., 2023. Enhancing U-net extraction of hydrographic features from IfSAR data in Alaska using shallow water channel depth models. 2023 International Cartographic Conference, August 13-18, 2023, Capetown, South Africa.
- Shavers, E., Stanislawski, L., Schott, J., Brosseau, Z., 2023. Automated mapping of culverts, bridges, and dams. 2023 International Cartographic Conference, August 13-18, 2023, Capetown, South Africa.
- Kronenfeld, B., Stanislawski, L., Buttenfield, B., Shavers, E.. 2023. Generalization quality metrics to support multiscale mapping: Hausdorff and average distance between polylines. ICA Commission on Generalization and Multiple Representation 1st CartoAI Workshop, Sept. 12, 2023, Leeds, UK.
- Stanislawski, L., Jaroenchai, J., Wang, S., Shavers, E., Duffy, A., Thiem, P., Jiang, Z., Camerer, A., 2023. Transferring deep learning models for hydrographic feature extraction from IfSAR data in Alaska. ICA Commission on Generalization and Multiple Representation 1st CartoAI Workshop, Sept. 12, 2023, Leeds, UK.
- Stanislawski, L., Jaroenchai, J., Shavers, E., Thiem, P., Jiang, Z., Wang, S., Camerer, A., 2023. Transferring deep learning knowledge for scaling up hydrographic feature extraction. U.S. Geological Survey Hydrography Community Call, September 26, 2023.

- Shavers, E.J., Stanislawski, L.V., Buttenfield, B.P., and Kronenfeld, B.J., Abbott, E. 2024. Rethinking sinuosity measurement and its relation to hydrologic and geomorphic characteristics. American Geophysical Union Chapman Conference on Remote Sensing of the Water Cycle: Sensors to Science to Society. Feb. 13-16, 2024, Honolulu, HI.
- Stanislawski, L.V., Kronenfeld, B.J., Buttenfield, B.P., Shavers, E.J., Camerer, A. 2024. Assessment of density pattern retention for generalized data. American Association of Geographers 2024 Annual Meeting, April 16-20, 2024, Honolulu, HI.
- Kronenfeld, B.J,. Stanislawski, L.V., Buttenfield, B.P., Shavers, E., and Kang, Y., 2024. Exact computation of Hausdorff and average distance between polylines to support intelligent multiscale mapping. American Association of Geographers 2024 Annual Meeting, April 16-20, 2024, Honolulu, HI.
- Stanislawski, L.V., Shavers, E., Pastick, N., Thiem, P., Wang, S., Jaroenchai, N., Jiang., Z., Kronenfeld, B., Buttenfield, B. P., Camerer, A.., Adaptive fine-tuning for transferring a U-net hydrography extraction model using k-means. Cartography and Geographic Information Society (CaGIS) and the University Consortium for Geographic Information Science (UCGIS) 2024 Symposium. June 3-6, 2024, Columbus, OH.
- Kronenfeld, B.J., Buttenfield, B.P., Stanislawski, L.V., Shavers, E. 2024. Untangling the knots: a procedure for identifying discernibility conflicts on a cartographic line. Cartography and Geographic Information Society (CaGIS) and the University Consortium for Geographic Information Science (UCGIS) 2024 Symposium. June 3-6, 2024, Columbus, OH.
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UPCOMING

- **Journal Articles**
- Jaroenchai, N., Wang, S., Stanislawski, L., Shavers, Jiang, Z., Sagan, V., Usery, E.L. (in work). Transfer learning with convolutional neural networks for hydrological streamline detection.
- Kronenfeld, B.J., Buttenfield, B.P., Stanislawski, L.V., Shavers, E. (in work). Exact algorithm for calculating Hausdorff and Average Euclidian distance between polylines using an R-Tree Index.
- **Conference Sessions**
  - Artificial Intelligence for Watershed Analysis, American Water Resources Association (AWRA), the Universities Council on Water Resources (UCOWR), and the National Institutes for Water Resources (NIWR) 60th Anniversary Joint Water Resource Conference, Sept. 30-Oct. 2, 2024, St. Louis, MO. Organizers: Li, R., Shavers, E., and Stanislawski, L.
- **Accepted for Presentations**
- Stanislawski, L., Shavers, E., Pastick, N., Thiem, P., Wang, S., Jaroenchai, N., Jiang, Z., Kronenfeld, B., Buttenfield, B.P. , Camerer, A. Implementation of transfer learning to extract hydrography for

Alaska from IfSAR data. AWRA, UCOWR and NIWR 60th Anniversary Joint Water Resources. Sept. 30-Oct 1., St. Louis MO.

- Shavers, E., Stanislawski, L., Kronenfeld, B., and Buttenfield, B.P. Exploring the user of scale specific sinuosity metrics for terrain analysis.AWRA, UCOWR and NIWR 60th Anniversary Joint Water Resources. Sept. 30-Oct 1., St. Louis MO.

- **Meetings**

- Sept. 5, NGTOC CEGIS discussion of hydro extraction research

- **Software and Data**

- AK_workflow: workflow for automated download and generation of DTM, DSM, and ORI datasets for one or more Alaska HU12 watersheds, along with several elevation-derived layers. (Github repository available, additional documentation and testing still needed). Coordinated with HPC project.

- CONUS_workflow: similar to AK_workflow but for lidar data in development. Coordinated with HPC project.

- Geoflow: U-net workflow for predicting hydrography from lidar and IfSAR elevation data, which includes data preprocessing, sampling, sample visualization, training, prediction, validation. (Github repository available, further testing and documentation needed).

- GQAT: Generalization Quality Assessment Tools. Open source tools developed through CDI-funded project.